

Evaluating Large Language Model Systems (Demo)

Why Evaluation Is Hard

Unlike classical machine learning tasks with a single well-defined metric, evaluating an LLM-based system usually requires assessing several distinct qualities at once: factual correctness, relevance to the question asked, fluency, safety, and — for retrieval-augmented systems specifically — whether the answer is actually grounded in the retrieved evidence rather than the model's parametric knowledge.

A system can produce a fluent, confident-sounding answer that is entirely unsupported by its sources; conventional text-quality metrics will not catch this, which is why grounded systems need dedicated groundedness and citation-correctness checks.

Retrieval Metrics

Retrieval hit rate measures, across a labeled evaluation set, the fraction of questions for which at least one relevant chunk was present in the retrieved set. This isolates retrieval quality from generation quality: even a perfect language model cannot answer correctly if the necessary evidence was never retrieved.

Other common retrieval metrics include precision@k (what fraction of the top-k results are relevant) and recall@k (what fraction of all relevant chunks appear in the top-k).

Generation and Grounding Metrics

Citation correctness checks whether the citations attached to an answer actually correspond to passages that were retrieved and that support the claim being cited. Answer groundedness measures whether the factual content of the answer is supported by the retrieved evidence, as opposed to being invented or drawn from the model's general training knowledge.

Refusal accuracy measures whether the system correctly declines to answer (for example, returning 'insufficient evidence') exactly when the evidence genuinely does not support an answer, and conversely does not refuse when it does have sufficient evidence — both over-refusal and under-refusal are failure modes worth measuring separately.

Operational Metrics

Average response latency is a basic but important operational metric, since a system that is accurate but too slow is not usable in practice. Latency in a RAG system is typically dominated by the retrieval step (embedding the query, vector search) and the generation step (the LLM call), and should be measured end-to-end rather than per component when reporting a user-facing number.

Building an Evaluation Set

A robust evaluation set for a grounded RAG system typically includes: answerable questions with a known-correct answer supported by specific documents; unanswerable questions where the knowledge base genuinely lacks the needed information, to test refusal behavior; questions where two ingested documents contain contradictory information, to test whether the system surfaces the contradiction rather than silently picking a side; and adversarial prompt-injection cases embedded in documents, to test whether the system's behavior can be hijacked by document content.